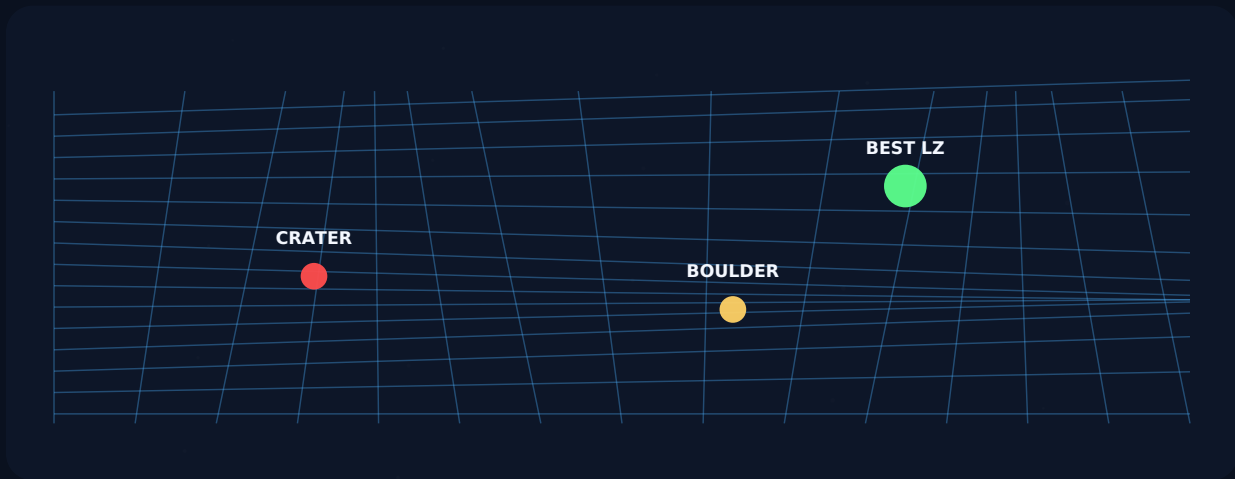


PLANETARY AUTONOMOUS LANDING ZONE ASSESSMENT

SLICK PDF USER GUIDE | PHYSICS-INFORMED SURVEY | HAZARD SCORING | ROVER PATH PLANNING

MISSION OPERATIONS CONSOLE



MISSION CONTROLS

Configure body, terrain, Sun, relay, scoring, rover, and optional LLM settings.

ANALYSIS ENGINE

Generates terrain, slope, hazard, illumination, line-of-sight, and LZ score maps.

PRIMARY OUTPUTS

3D surface, BEST LZ, heatmaps, habitat plan, rover route, JSON, and CSV.

AI BOUNDARY

Optional LLM output is advisory only. Physics remains authoritative.

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Space Age Theme

Conceptual Autonomy

Application Overview

WHAT THE APP DOES AND HOW TO INTERPRET IT

PURPOSE

This application performs a conceptual planetary landing-zone assessment for Moon, Mars, Phobos, Europa, or a custom airless body. It creates a synthetic terrain field, models crater and boulder hazards, evaluates slope, illumination, line-of-sight communication masking, and thermal penalty, then recommends the highest-scoring landing zone.

Use it as a trade-space and systems-engineering visualization tool. Do not present it as validated flight software, certified terrain-relative navigation, or a real planetary digital elevation model product.

ASSESSMENT FLOW

INPUTS

Mission ID
Target body
Terrain seed
Sun and relay geometry
Scoring weights



PHYSICS MAPS

Slope
Illumination
Terrain shadow
LOS communication
Hazard buffers



AUTONOMY LOGIC

Suitability score
Risk banding
A* rover route
Habitat offsets



EXPORTS

3D map
Heatmaps
Rover path
JSON report
CSV summary

BEST USE CASES

- > Compare target body presets and site quality trends.
- > Explain how mission inputs trace to visible outputs.
- > Generate repeatable screenshots and export files for documentation.
- > Support a LinkedIn or portfolio presentation with clear aerospace boundaries.

DO NOT CLAIM

- > Flight qualification or certification.
- > Real terrain validation from orbital data.
- > Operational landing authorization.
- > Certified autonomous hazard detection by the LLM.

Quick Start

A CLEAN WORKFLOW FROM SETUP TO EXPORT

FIVE-STEP RUN SEQUENCE

1. Open Mission Controls. 2. Select the target body and mission profile. 3. Tune terrain, illumination, communication, scoring, and rover settings. 4. Click Run Autonomous Assessment. 5. Review the LZ score, maps, habitat plan, rover path, recommendations, JSON, and CSV.

01

SELECT THE TARGET BODY

Choose Moon, Mars, Phobos, Europa, or Custom airless body. Each preset changes gravity, solar flux, atmosphere flag, slope limits, hazard buffer, communication mode, and thermal penalty.

02

BUILD THE SYNTHETIC TERRAIN

Set the seed, resolution, roughness, crater count, and boulder count. The seed makes the terrain repeatable for a report or portfolio screenshot.

03

SET MISSION GEOMETRY

Use Sun azimuth/elevation for illumination and shadow logic. Use relay or Earth azimuth/elevation for line-of-sight communication scoring.

04

RUN THE ASSESSMENT

The app computes slope, illumination, communication, hazard clearance, thermal score, landing suitability, risk band, and rover path.

05

EXPORT THE EVIDENCE

Export the JSON report and CSV summary after reviewing the 3D map, heatmaps, habitat plan, rover path, and recommendations.

Mission Controls

SIDEBAR INPUTS AND WHAT THEY CHANGE

MISSION SETUP

- > Mission ID labels the run for exports and reporting.
- > Target Body loads Moon, Mars, Phobos, Europa, or Custom.
- > Mission Profile frames the scenario: LZ, habitat, relay, or survey.

TERRAIN MODEL

- > Simulation Seed makes runs repeatable.
- > Map Resolution changes grid density.
- > Roughness, Crater Count, and Boulder Count shape the synthetic hazard field.

ILLUMINATION AND COMMS

- > Sun Azimuth/Elevation drive surface incidence and shadowing.
- > Relay or Earth Azimuth/Elevation drive terrain masking.
- > The caption shows gravity, solar flux, and communication mode.

SCORING WEIGHTS

- > Hazard Avoidance Weight emphasizes crater and boulder clearance.
- > Illumination Weight emphasizes solar and shadow quality.
- > Communication Weight emphasizes line-of-sight availability.

ROVER PLANNING

- > Nominal Rover Speed is adjusted by a gravity-based mobility factor.
- > Survey Radius and Waypoints define the route envelope.
- > The route uses hazard-aware cost mapping and A* planning.

OPTIONAL LLM ADVISORY

- > Disabled by default.
- > Requires OPENAI_API_KEY in Streamlit secrets or environment variables.
- > Generates advisory mission reasoning only. Physics remains authoritative.

Scoring Engine

HOW THE RECOMMENDED LANDING ZONE IS SELECTED

CORE SCORING LOGIC

The app combines hazard clearance, slope quality, illumination, communication visibility, and thermal penalty into a normalized 0 to 100 landing-zone suitability score. The recommended LZ is the grid cell with the highest suitability under the selected target body and input assumptions.

A high score means favorable simulated conditions. It does not certify a real landing site.

CONCEPTUAL SCORE COMPOSITION

**LZ Score = weighted hazard clearance + slope + illumination
+ communication line of sight + thermal margin**

Normalized to 0-100 after target-body limits and user-selected weights are applied

HAZARD CLEARANCE

Crater and boulder penalties are buffered by the selected body-specific safe hazard distance.

SLOPE SCORE

Terrain slope is compared against soft and hard slope limits for the selected target body.

ILLUMINATION

Sun vector, surface normals, terrain shadowing, and solar flux create an illumination index.

COMMUNICATIONS

Relay or Earth geometry is checked for terrain masking and line-of-sight availability.

THERMAL PENALTY

A body-level thermal factor lowers suitability where thermal risk is higher.

RISK BAND

Score ≥ 85 is Low, score ≥ 70 is Moderate, and below 70 is High.

Reading the Outputs

WHAT EACH PANEL MEANS

TOP METRICS

Fast read: LZ Score, Slope at LZ, Hazards Near LZ, Illumination Index, and Comm Index.

3D TERRAIN MAP

Main Plotly surface showing relative elevation and landing suitability, with crater hazards, boulder hazards, and BEST LZ markers.

SLOPE HAZARD MAP

Terrain steepness in degrees. Use it to check whether the recommended LZ is near soft or hard slope limits.

ILLUMINATION MAP

Sun incidence plus shadowing and solar flux. Use it for solar power and thermal context.

COMM MAP

Line-of-sight relay quality after terrain masking. Low values suggest relay placement needs review.

MISSION OVERVIEW

Mission ID, target body, profile, body constants, and the risk band for the selected LZ.

HAZARD SCORING

Counts craters, boulders, buffered hazards near the LZ, hazard clearance, and terrain shadow fraction.

HABITAT PLAN

Conceptual coordinates for landing, habitat, power, and communications relay zones with rationale.

ROVER PATH

Hazard-aware rover survey path plus path length, effective speed, estimated survey time, and coverage.

Exports and LLM Advisory

MISSION REPORT JSON

Preserves mission ID, target body, timestamp, body constants, geometry inputs, recommended LZ, score components, hazard statistics, rover survey metrics, recommendations, model status, and AI claim boundary. Use it as the full assessment record.

MISSION SUMMARY CSV

Provides a single-row mission summary with key body, geometry, LZ, hazard, illumination, communication, and rover-planning values. Use it to compare multiple seeds, bodies, or scoring-weight trade studies.

OPTIONAL LLM HAZARD ADVISOR

When enabled and configured, the LLM receives a compact engineering packet with mission inputs, body constants, recommended LZ data, map statistics, rover route metrics, and the explicit statement that the physics engine remains authoritative.

Correct interpretation: the LLM helps phrase Top Hazards, Avoidance Actions, Sensor Tasking, and Confidence and Limitations. It should not be described as certified autonomous hazard detection or as proof of real terrain validation.

Fallback behavior: if the LLM fails or no key is available, the app returns a deterministic rule-based autonomy brief.

RECOMMENDED ARCHIVE BUNDLE

- App version or code snapshot.
- Target body, mission profile, and simulation seed.
- Exported JSON and CSV files.
- Screenshots of the 3D map, heatmaps, rover path, and recommendations.
- Clear statement that results are conceptual, physics-informed, and dependent on input assumptions.

Accuracy Boundaries

AEROSPACE WORDING THAT STAYS DEFENSIBLE

AEROSPACE ACCURACY BOUNDARY

This guide describes the application faithfully: a physics-informed conceptual simulation using synthetic terrain, simplified body constants, slope scoring, solar geometry, terrain shadowing, line-of-sight masking, hazard buffering, rover path planning, and optional advisory LLM reasoning.

It does not claim high-fidelity orbital mechanics, certified landing dispersion, validated sensor performance, real planetary DEM ingestion, plume-surface interaction modeling, formal verification, or mission assurance certification.

PRE-RUN CHECKLIST

- > Confirm target body and mission profile.
- > Record the simulation seed.
- > Set plausible crater and boulder density.
- > Review Sun and relay geometry.
- > Adjust scoring weights intentionally.
- > Leave LLM disabled unless advisory review is needed.

POST-RUN CHECKLIST

- > Check LZ score and risk band.
- > Verify slope at the selected LZ.
- > Inspect crater and boulder proximity.
- > Compare illumination and communication maps.
- > Review rover path length and survey time.
- > Export JSON and CSV for documentation.

SUGGESTED PRESENTATION LANGUAGE

Strong wording: "This application demonstrates a conceptual planetary landing-zone assessment workflow using physics-informed terrain scoring, synthetic hazard fields, illumination modeling, line-of-sight communication masking, and rover survey planning."

Avoid wording: "This app certifies safe landing sites" or "The LLM performs flight-qualified autonomous hazard detection."

LinkedIn framing: conceptual autonomy, transparent assumptions, exportable mission summaries, and systems-engineering traceability from inputs to outputs.